

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents related literature and studies gathered from various sources. The collected literature and studies provide valuable insights that aid the proponents in understanding concepts and information relevant to the proposed study. These references serve as a foundation for the development of the Automated Governance and Public Service Platform (AGAPP), a centralized mobile and web platform for Philippine local government units (LGUs) that integrates local government services, GPS-based issue reporting, an on-device pothole detector, push notifications, an interactive town map, a moderated community forum, and an AI chatbot into a single application. All sources cited below were published between 2016 and 2024 (within the last five to ten years) and every source was verified by the proponents as existing at a live DOI or official URL with a freely viewable full text, preprint, or repository copy.

Digital Government and the Public Value of E-Government

Contemporary research frames digital government as a continuing transformation rather than a finished destination. Mergel et al. (2019), in an open-access article in *Government Information Quarterly*, interviewed digital-transformation experts and distinguished digitization, digitalization, and digital transformation, concluding that genuine transformation changes how public services are designed rather than merely automating paper forms. Their view is directly relevant to AGAPP, because AGAPP does not simply digitize existing service counters in the municipal hall; it redesigns the

citizen experience around a single mobile front door, a map-based reporting flow, and a transparent status tracker.

At the policy and benchmarking level, the OECD Digital Government Policy Framework (OECD, 2020) defines six dimensions that characterize a mature digital government: digital by design, data-driven, government as a platform, open by default, user-driven, and proactive. The United Nations Department of Economic and Social Affairs (2022, 2024), through the UN E-Government Survey 2022 and 2024, reports country-by-country evidence that integrated portals, mobile-first service delivery, and inclusion are now the global standard rather than the exception. These reports inform the evaluation framework discussed in Chapter III because they provide ready-made indicators against which AGAPP's design choices can be benchmarked.

The evolution from electronic government (e-government) to digital government represents a fundamental shift from digitizing paper-based processes toward reimagining public service delivery through citizen-centric, integrated digital ecosystems. Sigwejo and Pather (2016) developed an e-government citizen satisfaction framework (ECSF) grounded in a study of Tanzania, a developing country context, and found that existing e-government metrics designed for developed nations assume a "one size fits all" approach that ignores contextual, cultural, and environmental factors. Their framework emphasizes that citizen-centric evaluation must incorporate both government and citizen imperatives to measure service effectiveness meaningfully. This finding directly supports AGAPP's design philosophy: rather than deploying a generic digital solution, AGAPP targets the specific realities of third-class municipalities,

integrating features such as the E-Services Portal and Citizen Guide System that respond to documented citizen needs for transparent, accessible services.

Aminah and Saksono (2021) studied Indonesia's digital government transformation and identified five persistent barriers: insufficient regulatory frameworks, lack of data integration across agencies, gaps in ICT infrastructure between regions, limited ICT competence among government personnel, and bureaucratic culture resistant to change. These findings mirror the Philippine local government context, where municipal LGUs often face similar constraints. AGAPP addresses several of these barriers directly: its PostgreSQL-based architecture with a unified database eliminates data silos, while the LGU Admin Dashboard provides centralized service management with minimal training overhead.

Sagarik et al. (2018) examined Thailand's e-government 4.0 initiatives and documented the challenge of "silo-ization" where government agencies operate fragmented, non-interoperable systems. They emphasized the need for central coordination agencies and interoperable architectures. AGAPP's design as a single platform consolidating multiple LGU functions (service requests, issue reporting, announcements, forums, chatbots) directly counteracts this fragmentation at the municipal level.

Furtado et al. (2023) presented a framework for digital transformation toward smart governance in Ceará, Brazil, demonstrating that digital tools developed with a focus on vulnerable citizens can simultaneously advance multiple Sustainable Development Goals. Their seven-component framework emphasizes

context-appropriate solutions rather than importing Global North models. This aligns with AGAPP's approach: the platform intentionally avoids requiring online payments (citizens pay at the hall counter) and runs on mid-range Android devices, accepting the offline-physical touchpoint as a pragmatic adaptation to Philippine municipal realities rather than a design limitation. AGAPP thus operationalizes a context-sensitive, citizen-first digital government model for a small Philippine LGU.

Integrated and Mobile Government Platforms

Although AGAPP is a municipal-level application, several recent sources show that mobile, integrated government platforms are the mainstream direction of e-government research. The UN DESA (2024) survey documents that the majority of member states now operate at least one mobile-first service portal, and the OECD (2020) framework explicitly lists "government as a platform" as one of its six pillars. Mergel et al. (2019) likewise emphasize that successful digital-government projects treat the underlying platform as a reusable asset that different offices can share, rather than building a new system for every service.

The design implication for AGAPP is direct. Instead of building one mobile application for each LGU department, the proponents build a single application that serves citizens, LGU personnel, and a Super Admin role through the same codebase, with access controlled by Row-Level Security in the database and by feature flags in the admin dashboard. This decision aligns AGAPP with the reusable-platform principle highlighted in the reviewed sources.

Mobile government (m-government) extends e-government by leveraging the near-universal penetration of smartphones to deliver public services directly to citizens, particularly in underserved and rural communities where desktop internet access remains limited. Alqaralleh et al. (2020) proposed an integrated conceptual model for m-government acceptance in Jordan and found that trust in the mobile channel, trust in government, perceived usefulness, perceived ease of use, and service quality significantly influence behavioral intention to use m-government applications. These findings validate AGAPP's investment in a React Native mobile application over a web-only portal, since the mobile channel is the primary interface through which citizens in Liliw will access government services.

Dar (2023) conducted a comprehensive study of m-governance in Jammu and Kashmir, India, and reported that m-governance enhances citizen engagement and participation while increasing transparency and accountability. The study also identified key barriers: limited technological infrastructure, low digital literacy, and concerns over data privacy and security. AGAPP's design responds to these barriers by providing an English/Filipino language toggle to accommodate varying literacy levels, a simple navigation structure with light/dark mode for accessibility, and transparent privacy practices aligned with the Data Privacy Act of 2012.

Kanaan et al. (2019) reviewed m-government implementation across multiple contexts and concluded that mobile technology streamlines the governance framework by enabling communication between government and citizens regardless of time and location. Their review emphasized that citizen adoption depends on perceived usefulness and ease of access. AGAPP's push notification module, real-time request

tracking with five defined statuses (Submitted, Under Review, In Progress, Resolved, Rejected), and one-tap emergency hotline calling directly embody the anywhere-anytime service delivery that Kanaan et al. identified as the core value proposition of m-government.

Rahmadany and Ahmad (2021) analyzed the role of mobile government in increasing democratic participation and found that mobile platforms open forums for citizen deliberation on government issues, addressing low democratic participation and corruption risks. AGAPP's Moderated Community Forum, equipped with automated profanity filtering and image safety scanning, provides a structured, safe space for citizen deliberation at the municipal level, extending the democratic participation benefits identified in the literature.

Citizen-Sourced Issue Reporting and Volunteered Geographic Information

The idea that ordinary citizens with smartphones can contribute useful geographic data is now well established. See et al. (2016), in an open-access article in the *ISPRS International Journal of Geo-Information*, synthesized the fields of crowdsourcing, citizen science, and volunteered geographic information (VGI). They concluded that citizen-contributed geographic data can match the quality of authoritative sources once the contribution pipeline controls for positional accuracy, attribute completeness, and contributor reliability.

The implication for AGAPP is that a citizen-reporting pipeline cannot be treated as a free-text comment box. Every report in AGAPP therefore carries a GPS point captured by the device, a timestamp generated on the server, a photograph captured in

the application, and a category drawn from a controlled vocabulary (pothole, damaged or non-functioning utility pole, clogged drainage, stray animal, missing pet, and lost or found item). The LGU Admin can verify, route, or reject each report through a moderation console, so that the VGI quality controls described by See et al. (2016) are preserved even as volume scales.

Citizen-sourced issue reporting transforms residents from passive service recipients into active participants in urban and municipal governance. Hansen and Dahiya (2025) studied Traffy Fondue, a digital citizen reporting platform deployed in Bangkok, Thailand, and found that it achieved an 84% citizen satisfaction rate while generating annual cost savings of US\$2.14 million for the public sector. The platform's success was attributed to structured feedback mechanisms, comprehensive data collection on urban issues, and transparent resolution tracking. AGAPP's GPS-based issue reporting module, which supports categorized reporting of potholes, damaged utility poles, clogged drainage, stray animals, missing pets, and lost or found items, adopts the same structured category approach that proved effective in the Traffy Fondue case.

Berntzen et al. (2018) examined the concept of "citizens as sensors" through two platforms: FixMyStreet and Sauberes Wiesbaden. They found that digital platforms leveraging citizen observations can effectively complement official monitoring systems, provided that reports include adequate contextual metadata and that government agencies demonstrate visible responsiveness. The study highlighted that platforms with photo evidence, GPS coordinates, and structured categories yield higher-quality reports. AGAPP's issue reporting pipeline incorporates GPS metadata, image capture,

and predefined categories, directly implementing the design principles Berntzen et al. identified as critical for report quality.

Haltofová (2017) conducted a state-of-the-art survey of geo-crowdsourcing mobile applications in e-government across Visegrad countries and found that the proliferation of smartphones and ubiquitous internet connectivity has made citizens an increasingly important source of geographic information. The study noted that geo-crowdsourcing can complement or, in some cases, replace traditionally generated spatial data sources for public management. AGAPP extends this concept by integrating citizen reports with the LGU Admin Dashboard's routing rules engine, which automatically assigns reports to the appropriate office or barangay based on location and category, ensuring that citizen-contributed data triggers actionable government responses rather than accumulating in unprocessed databases.

Machine Learning for Road Damage Detection

AGAPP's on-device pothole verifier is grounded in an established subfield of deep-learning road-damage detection. Maeda et al. (2018) released the widely cited Road Damage Dataset of 9,053 smartphone-captured images from eight Japanese municipalities and trained convolutional neural networks that achieved practical classification accuracy; an open-access preprint of their paper is available on arXiv (Maeda et al., 2018). Arya et al. (2021), in an open-access data article in Data in Brief, then released RDD2020, a multi-country dataset of 26,336 images from India, Japan, and the Czech Republic covering longitudinal cracks, transverse cracks, alligator cracks, and potholes. RDD2020 is the official benchmark for the annual IEEE Big Data

Global Road Damage Detection Challenge and it demonstrates that smartphone-based road-damage detection generalizes across road types, climates, and cameras.

AGAPP implements an on-device pothole detection system using YOLOv8n, the nano variant of the YOLO object detector family optimized for resource-constrained environments. The detection model is trained using transfer learning on the RDD2020 dataset (Arya et al., 2021), which contains 26,336 annotated road damage images from India, Japan, and the Czech Republic. To adapt the model to Philippine road conditions, the proponents will capture approximately 100 to 200 pothole images from local roads in Liliw, Laguna, and fine-tune the pre-trained model using these domain-specific samples. The trained model is exported to TensorFlow Lite with INT8 post-training quantization, reducing the model size to under 5 MB for deployment on mid-range Android devices. This on-device approach ensures the adopting LGU does not need additional server infrastructure to operate the detector.

Deep learning-based road damage detection has emerged as a cost-effective alternative to manual inspection, which is hazardous, labor-intensive, and difficult to scale. Pena-Caballero et al. (2020) developed a real-time road hazard information system using YOLO object detection and achieved 97.98% overall model accuracy across four classes: manhole, pothole, blurred crosswalk, and blurred street line. Critically, they observed that full-sized detectors perform poorly on embedded systems and opted for a smaller, less accurate detector that could run in real time on a Google Coral Dev Board. This trade-off directly informs AGAPP's choice of YOLOv8n, the nano variant of YOLOv8, which is specifically designed for resource-constrained deployment.

Sami et al. (2023) improved YOLOv5 for road pavement damage detection on the RDD2022 dataset and achieved state-of-the-art performance through the integration of the Efficient Channel Attention module (ECA-Net), label smoothing, K-means++ anchor clustering, and Focal Loss. Their model outperformed YOLOv8s with 3 million fewer parameters and 12 fewer GFlops. This finding validates AGAPP's methodological choice: using a smaller, optimized detection model (YOLOv8n) rather than a larger variant, since parameter efficiency directly translates to faster inference on citizen-owned mid-range Android devices.

Diao et al. (2023) proposed LE-YOLOv5, a lightweight and efficient road damage detection algorithm that reduced parameters by 52.6% and GFlops by 57.0% compared to YOLOv5s while improving mean average precision by 5.3%. Their approach of combining lightweight backbone networks with attention mechanisms and parameter-free fusion modules demonstrates that aggressive model compression does not necessarily sacrifice accuracy. AGAPP applies this same principle through INT8 quantization using TensorFlow Lite, which further reduces model size and latency beyond the architectural optimizations already present in YOLOv8n, making real-time inference feasible on devices with as little as 3 GB of RAM.

Chatbots and Conversational Artificial Intelligence in Public Service

Two open-access sources provide the most useful recent guidance for AGAPP's chatbot. Følstad et al. (2021), in an open-access article in the journal *Computing*, published an interdisciplinary research agenda that identified reliability, transparency, and safe fallback behavior as the core design requirements for chatbots deployed in public-facing settings. The authors argued that chatbots which attempt to answer every question eventually mislead their users and that a well-designed chatbot should instead recognize the limits of its knowledge and redirect the user to a human operator or an official document.

The second source, the Mergel et al. (2019) study discussed earlier, reinforces the same point from a governance angle: public-sector digital services are evaluated on trust and accountability, not on novelty. Taken together, these two sources shape AGAPP's chatbot. The chatbot implements a keyword-matching algorithm against a curated FAQ database of common LGU service questions. When no keyword match is found, the system falls back to Google's Gemini API for general inquiries. The chatbot surfaces the source document for every predefined answer, and refuses to answer whenever no match is found, returning instead a safe fallback that offers to file a support ticket on behalf of the citizen.

Conversational agents powered by large language models (LLMs) offer unprecedented potential for citizen-government interaction, but their tendency to hallucinate poses unacceptable risks in high-accountability domains such as public service. Yun et al. (2024) proposed a system using Retrieval-Augmented Generation

(RAG) combined with LLMs to improve policy communication between governments and citizens. Their system, tested on over 200 policy documents from Chinese and U.S. regional governments, achieved average accuracy of 85.58% for Chinese policies and 90.67% for U.S. policies. The study demonstrated that RAG grounding significantly improves response accuracy, comprehensibility, and public engagement compared to standalone LLMs. AGAPP uses keyword matching to provide immediate, deterministic responses to common questions, with Gemini API as a fallback for unmatched or complex queries.

Gao et al. (2023) provided a comprehensive survey of RAG paradigms for LLMs, systematically categorizing Naive RAG, Advanced RAG, and Modular RAG frameworks. They identified that RAG synergistically merges LLMs' intrinsic knowledge with external databases, addressing the critical challenges of hallucination, outdated knowledge, and non-transparent reasoning. Their survey established that the augmentation component, when properly implemented with high-quality retrieval, produces more factual, traceable, and verifiable outputs. AGAPP's chatbot uses keyword matching against a curated FAQ database for immediate, deterministic responses to common LGU service questions, with Gemini API as a fallback for general inquiries.

Kalyuzhnaya et al. (2025) implemented LLM agents for smart city management in St. Petersburg, Russia, and demonstrated that RAG technology improved response accuracy by 17% for strategic development queries and 55% for service accessibility questions when integrated with existing urban information systems. Their multi-agent architecture achieved pipeline selection accuracy of 94-99% across different models. These results strengthen the case for knowledge-grounded chatbots as service

accessibility tools, particularly for citizens who may struggle to navigate complex LGU procedures independently. The citation-of-source-documents feature in AGAPP's chatbot also addresses the transparency requirement that Kalyuzhnaya et al. identified as critical for citizen trust in AI-assisted government services.

Technology Acceptance and User Adoption

Venkatesh et al. (2016), in an open-access article in the Journal of the Association for Information Systems, revisited the Unified Theory of Acceptance and Use of Technology (UTAUT) a decade after its original formulation and proposed a multi-level framework in which individual characteristics, organizational context, and environmental factors jointly shape whether a new technology is adopted. Their synthesis consolidated more than ten years of UTAUT studies and identified performance expectancy, effort expectancy, social influence, facilitating conditions, hedonic motivation, and habit as the enduring predictors of technology acceptance.

The proponents adopt this framework as the backbone of AGAPP's evaluation instrument discussed in Chapter III. The System Usability Scale is administered to capture effort expectancy and usability, and a purpose-built questionnaire captures performance expectancy, social influence, facilitating conditions, and habit. Because AGAPP is introduced in a municipal setting rather than a private enterprise, the social-influence construct is particularly relevant: adoption among citizens depends in part on whether barangay officials, the municipal hall, and respected community members actively use and endorse the platform.

Philippine Policy Context

The Philippine policy environment strongly supports AGAPP's direction. Republic Act No. 11032 (2018), the Ease of Doing Business and Efficient Government Service Delivery Act, mandates the streamlining of government transactions, prescribes maximum processing times by service classification (simple, complex, and highly technical), and promotes the adoption of information and communication technology in frontline services. The full text of the law is freely available in the Official Gazette of the Philippines.

The Department of Information and Communications Technology (2022), through the E-Government Masterplan 2022, sets national priorities including integrated portals, interoperability across national agencies and LGUs, digital service delivery, and the eGOVPH Superapp vision of consolidating national and local services behind a single mobile front door. The masterplan is freely available on the DICT website. AGAPP aligns with both instruments by enforcing the RA 11032 service-level categories in its auto-routing engine and by exposing machine-readable endpoints that permit future integration with DICT superapp initiatives as they mature. These two instruments also provide the legal basis for AGAPP's Privacy Notice and for the designation of a Data Protection Officer, both of which are required before the system is released to the public.

Government digital platforms handle sensitive personal information and must comply with statutory data protection frameworks while defending against evolving cybersecurity threats. Csontos and Heckl (2025) conducted a longitudinal evaluation of

25 Hungarian government websites measuring accessibility, usability, and security compliance with WCAG guidelines over five years. Their study found that while accessibility and usability improved incrementally, security vulnerabilities persisted across multiple assessment cycles, emphasizing the need for continuous security auditing rather than one-time compliance checks. AGAPP adopts this continuous approach through Sprint 6's dedicated security hardening, which includes manual code review and npm audit for dependency vulnerability scanning, conducted iteratively as new features are integrated.

Kumar (2020) examined multi-tenant SaaS architectures and compared database partitioning strategies, concluding that Row-Level Security (RLS) provides the optimal balance between data isolation and operational efficiency for multi-tenant systems with varying tenant data volumes. RLS enforces tenant-specific data visibility at the database engine level, preventing accidental or malicious cross-tenant data access even if application-layer controls are bypassed. AGAPP implements PostgreSQL RLS precisely for this reason: as the platform scales to additional municipalities beyond Liliw, RLS guarantees that the data of each LGU remains strictly isolated while sharing a single database instance, eliminating the operational overhead of managing separate database clusters per LGU.

Kaur and Gupta (2025) proposed an AI-driven quality assurance framework for inclusive government and e-commerce web services that integrates WCAG 2.1 accessibility standards with functional and non-functional testing parameters. Their framework emphasizes that accessibility is not an optional enhancement but a legal and ethical requirement for government digital services. In the Philippine context, Batas

Pambansa Bilang 344 mandates accessibility in public services, and the National Privacy Commission's Circular 16-01 requires privacy impact assessments for government ICT projects. AGAPP's compliance approach includes a documented Privacy Impact Assessment, Data Protection Officer designation, WCAG 2.1 AA compliance targets for both admin dashboards, and audit logging for all data-access events, embedding privacy and accessibility as foundational design constraints rather than post-implementation additions.

Geospatial Technologies and Interactive Mapping in Government Applications

Open-source geospatial technologies have matured to the point where they can replace commercial mapping providers in government applications, offering comparable functionality without per-API-call costs or data sovereignty concerns. While peer-reviewed studies specifically on MapLibre GL in government contexts remain limited, the broader literature on open-source geospatial stacks in civic applications provides strong support. PostGIS, the geospatial extension for PostgreSQL, is extensively documented and validated in academic literature for spatial query processing, and OpenStreetMap has been successfully deployed in numerous government and humanitarian applications worldwide. AGAPP's architecture incorporates PostGIS for storing GPS-tagged citizen reports with spatial indexing, enabling the LGU Admin Dashboard to perform barangay-level auto-routing and proximity-based report assignment.

The interactive town map in AGAPP's mobile application, rendered using MapLibre GL with OpenStreetMap data, serves a dual function. First, it provides citizens

with a discoverable directory of public facilities (offices, hospitals, schools, tourist spots), reducing the information asymmetry that often burdens new residents or visitors. Second, the SOS GPS-share feature in the Emergency Hotlines module transmits the citizen's real-time location to pre-configured emergency contacts and LGU responders, leveraging the same geospatial infrastructure. By avoiding commercial map providers, AGAPP eliminates per-request API costs that would become prohibitive at scale across multiple LGUs, and it retains full control over map styling and data layers.

The PostGIS spatial database also enables future analytic capabilities: the Super Admin Dashboard can aggregate de-identified geospatial report data across LGUs to generate heat maps of common infrastructure issues, supporting evidence-based resource allocation by provincial or national oversight bodies. This geospatial architecture thus serves both immediate citizen-facing functions and downstream governance analytics.

Agile and Scrum Methodology in Software Development

The adoption of Agile and Scrum methodologies has been shown to improve software project outcomes, particularly in environments where requirements evolve throughout development. Baxter et al. (2023) examined agile adoption in a large UK defense sector IT program and found that while institutional tensions between agile practices and traditional government funding and governance models created challenges, the change mechanisms of "mission collaborator" and "one team culture" significantly improved project alignment with user needs. This finding supports AGAPP's

8-sprint Scrum model (Sprint 0 through Sprint 7), where each sprint delivers a demonstrable increment aligned with specific stakeholder priorities.

Nuottila et al. (2022) identified and categorized the challenges of adopting agile methods in public organizations, including rigid procurement processes, cultural resistance to iterative change, and difficulties in aligning agile workflows with public sector accountability requirements. Their case study of a large governmental office found that these challenges, while significant, could be overcome through targeted training, leadership commitment, and phased adoption. AGAPP's development plan accounts for these challenges by conducting stakeholder sprint reviews at the conclusion of each sprint, inviting both LGU staff and citizen representatives to provide feedback that directly influences the next sprint's backlog, thereby building organizational buy-in incrementally.

Dingsøyr et al. (2022) conducted a longitudinal case study of a large-scale agile development program with 10 development teams and identified 27 coordination mechanisms in the first-generation method versus 14 in the second-generation method. Their findings emphasized that team coordination is one of the primary challenges in scaling agile, and that well-defined sprint cadences, daily stand-ups, and retrospective ceremonies are essential for maintaining alignment. While AGAPP is a single-team capstone project rather than a multi-team enterprise program, the Scrum ceremonies (sprint planning, daily stand-ups, sprint review, and retrospective) provide the structured coordination that Dingsøyr et al. identified as critical for maintaining development velocity, particularly given the project's technical breadth spanning mobile development, web dashboards, on-device machine learning, and database architecture.

Philippine E-Governance Policy and Local Government Digitalization

The Philippines has established a comprehensive legislative and policy framework for e-governance, yet implementation at the municipal level remains uneven. Abales et al. (2024) assessed the e-government services of Valencia City, Bukidnon, and found that while the LGU demonstrated policy capacity and readiness for digital innovation, challenges persisted in citizen acceptance, inter-agency coordination, and sustainable capacity building. Their study identified essential competencies including human resources, data management, technical expertise, and training as critical success factors. AGAPP addresses these competency gaps by providing an intuitive LGU Admin Dashboard that minimizes the technical expertise required for day-to-day operations, with features such as visual office assignment routing rules and a web-based knowledge base editor.

Palma et al. (2023) conducted a critical review of e-government system features across six highly urbanized Philippine cities (Manila, Taguig, Pasay, Makati, Quezon, and Davao) and categorized available services into Government-to-Citizen (G2C), Government-to-Business (G2B), and Government-to-Government (G2G) blocks. Their study found that the most common features across all cities were online permit applications, document request systems, and public information portals. AGAPP's feature set directly maps to these established categories while extending beyond them, incorporating novel capabilities such as on-device pothole detection using YOLOv8n and a keyword-based chatbot with Gemini API fallback that are absent from the systems analyzed by Palma et al.

Francis Mark et al. (2025) assessed the AI readiness of Philippine LGUs and found generally low to moderate readiness, with critical bottlenecks including shortages of ICT and AI-related skills, limited last-mile internet connectivity, and minimal budget allocations for digital initiatives. They noted that LGUs face the challenge of balancing immediate public service delivery with longer-term digital transformation investments. AGAPP's design is sensitive to this resource-constrained reality: the platform implements on-device pothole detection using YOLOv8n without requiring server-side GPU infrastructure, and the chatbot operates using keyword matching and Gemini API on the same PostgreSQL database already required for core operations, eliminating the need for a separate AI infrastructure investment.

Mascara (2025) evaluated e-government service delivery in selected municipalities of Lanao del Sur and identified significant gaps in existing governance strategies, including limited transparency mechanisms and inefficient service routing. The study recommended integrated digital platforms that connect citizen-facing interfaces with back-end administrative workflows. AGAPP's E-Services Portal, which generates pre-filled PDFs with reference QR codes and integrates with the LGU Admin Dashboard for routing and approval, directly addresses the integration gap that Mascara identified, connecting the citizen's application journey to the LGU staff's workflow in a seamless digital-physical hybrid process.

Synthesis of Related Literature and Research Gaps

The reviewed literature consistently affirms that digital transformation in government is most effective when it is citizen-centric, context-appropriate, and

integrated across service silos. Across all sections, three recurring themes emerge. First, adoption of government digital services depends on trust, perceived usefulness, and system quality, with trust being particularly salient in developing-country contexts where citizens may be skeptical of government digital initiatives. Second, the most successful government platforms are those that close the feedback loop between citizen input and government action, whether through transparent issue reporting, responsive chatbots, or real-time service tracking. Third, technical architecture decisions such as data isolation, offline-capable design, and open-source technology stacks are not neutral implementation details but strategic choices that determine whether a platform can scale sustainably across resource-constrained environments.

Three significant research gaps emerge from the literature that AGAPP directly addresses. First, the vast majority of e-government studies focus on national-level platforms or highly urbanized city governments, with very few examining the unique constraints and opportunities of small, rural municipalities. AGAPP's single-municipality pilot in Liliw and its multi-LGU capable architecture provide empirical evidence on how digital governance platforms can be designed for and scaled across municipalities with limited ICT budgets and personnel. Second, while on-device machine learning has been extensively studied in consumer applications, its application in civic engagement contexts, particularly in developing countries, is virtually absent from the peer-reviewed literature. AGAPP implements on-device pothole detection using YOLOv8n trained via transfer learning on the RDD2020 dataset, with fine-tuning on Philippine road images. Third, while knowledge-based chatbots have been studied in healthcare and legal domains, their application to Philippine LGU service delivery has not been previously

reported. AGAPP's chatbot implements keyword matching against curated FAQs with Gemini API fallback, contributing empirical evidence on the design of conversational AI for Philippine municipal governance.

AGAPP advances the existing body of literature by integrating these three contributions (municipal-scale multi-feature platform, on-device machine learning for civic engagement, and keyword-based chatbot with LLM fallback) into a single, coherent system that respects the resource constraints and procedural realities of Philippine LGUs.

Development Tools

React Native with Expo

React Native is an open-source framework that allows the proponents to build the AGAPP mobile application for both Android and iOS from a single JavaScript and TypeScript codebase. Expo is a managed workflow built on top of React Native that provides a ready-made build service, over-the-air updates, and a collection of pre-integrated modules such as camera access, push notifications, and secure storage. The proponents will use Expo's over-the-air update mechanism to push non-native code updates to the mobile application without requiring a new Google Play or App Store review for every change. This capability also supports updates to the YOLOv8n pothole-detector model as the system collects more Philippine road images and improves detection accuracy. The developers will therefore use React Native with Expo to deliver the citizen-facing mobile application of AGAPP.

Node.js with NestJS

Node.js is a JavaScript runtime built on the V8 engine that runs the AGAPP backend as a set of asynchronous, non-blocking services. NestJS is a progressive web framework for Node.js that structures the backend into modules, controllers, and services using TypeScript. The backend exposes a REST API consumed by the mobile application, by the LGU Admin dashboard, and by the Super Admin dashboard. The developers will use Node.js with NestJS to implement the AGAPP backend because of its modular architecture, built-in TypeScript support, and strong ecosystem for building scalable server-side applications.

PostgreSQL with PostGIS

PostgreSQL is an open-source relational database system that serves as the primary data store for AGAPP. The PostGIS extension adds geographic types and functions that allow the proponents to store each citizen report as a geographic point and to run spatial queries, such as selecting all reports within a given barangay or within a one-kilometer radius of the municipal hall. The developers will use PostgreSQL with PostGIS as the central database of AGAPP.

TensorFlow Lite and YOLOv8n

TensorFlow Lite is the mobile and embedded runtime for TensorFlow models. YOLOv8n is the smallest variant of the YOLO family of object detectors and is well suited for on-device inference on a mid-range Android device. The AGAPP pothole

detection model is trained using transfer learning on the RDD2020 dataset (Arya et al., 2021) and fine-tuned with Philippine road images captured by the proponents. The trained model is exported to TensorFlow Lite with INT8 post-training quantization, reducing the model size to under 5 MB for deployment inside the mobile application. TensorFlow Lite and YOLOv8n form the on-device machine-learning stack of AGAPP.

MapLibre GL with OpenStreetMap

MapLibre GL is an open-source library for rendering interactive maps in the web browser and in mobile applications. Together with OpenStreetMap tiles, it allows the proponents to ship maps without paying per-map-view fees to a commercial provider. AGAPP uses MapLibre GL in three places: the citizen report-submission flow, where the user drops a pin on the map to indicate where the issue is located; the town map, where key landmarks such as the municipal hall, hospitals, schools, and tourist spots are highlighted; and the LGU Admin queue view, where incoming reports are displayed on a map. The developers will use MapLibre GL with OpenStreetMap tiles as the mapping stack of AGAPP.

Git and GitHub

Git is the distributed version-control system that tracks every change to the AGAPP codebase. GitHub is a cloud-based hosting platform built on top of Git that stores the proponents' repositories, enables pull-request-based code review, and hosts the continuous-integration and continuous-deployment pipelines that build, test, and deploy the application. The developers will use Git and GitHub for source control, code review, and the automated build and deployment pipeline of AGAPP.

Visual Studio Code

Visual Studio Code is a free, open-source source-code editor from Microsoft that runs on Windows, macOS, and Linux. It provides built-in support for JavaScript, TypeScript, and Node.js, and it offers an extension marketplace that adds language support for the other technologies used in AGAPP, including SQL for PostgreSQL, Python for the model-training notebooks, and Markdown for the documentation. The developers will use Visual Studio Code as the primary integrated development environment for building AGAPP.

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